

EXPERIMENT NO:9
DATE:02-08-2026
COURSE: ARTIFICIAL INTELLIGENCE
CODE: CSA1715

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Write a Python program to implement Travelling Salesman

Problem

Aim:

To find the shortest possible route that visits each city exactly once and returns to the starting city.

Algorithm:

1. Represent cities using a distance matrix.
2. Start from a fixed city (say city 0).
3. Generate all possible permutations of remaining cities.
4. For each permutation:
 - a. Calculate total distance of the route.
 - b. Include return to starting city.
5. Compare all routes and select the minimum cost.
6. Print the shortest path and minimum distance.

Python Code:

```
import itertools

# Distance matrix
graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
```

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```
]
```

```
n = len(graph)
```

```
cities = list(range(n))
```

```
# Starting city
```

```
start = 0
```

```
min_path = float('inf')
```

```
best_route = []
```

```
# Generate permutations
```

```
for perm in itertools.permutations(cities):
```

```
    if perm[0] == start:
```

```
        current_cost = 0
```

```
        for i in range(n - 1):
```

```
            current_cost += graph[perm[i]][perm[i + 1]]
```

```
        # Return to start
```

```
        current_cost += graph[perm[-1]][start]
```

```
        # Print each route with explanation
```

```
        print("Route:", perm, "->", start)
```

```
        print("# Cost Calculation:", current_cost, "\n")
```

```
        if current_cost < min_path:
```

```
            min_path = current_cost
```

```
            best_route = perm
```

```
# Output
```

```
print("Shortest Route:", best_route, "->", start)
```

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```
print("Minimum Cost:", min_path)
```

Input:

Number of Cities = 4

Distance Matrix:

0 10 15 20

10 0 35 25

15 35 0 30

20 25 30 0

Output :

```
Route: (0, 1, 2, 3) -> 0  
# Cost Calculation: 95
```

```
Route: (0, 1, 3, 2) -> 0  
# Cost Calculation: 80
```

```
Route: (0, 2, 1, 3) -> 0  
# Cost Calculation: 95
```

```
Route: (0, 2, 3, 1) -> 0  
# Cost Calculation: 80
```

```
Route: (0, 3, 1, 2) -> 0  
# Cost Calculation: 95
```

```
Route: (0, 3, 2, 1) -> 0  
# Cost Calculation: 95
```

```
Shortest Route: (0, 1, 3, 2) -> 0  
Minimum Cost: 80
```

Result:

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The Travelling Salesman Problem is solved successfully, and the shortest route with minimum cost is obtained.